

Math 4102: Introduction to Lebesgue Integration

Final Examination Problems List

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Problem 1 (30.1, p.85)

State and prove Egoroff's theorem.

Proof.

□

Problem 2 (30.3, p.87)

State and prove Lusin's theorem.

Proof.

□

Problem 3

State the definition of a normed linear space and give both finite- and ∞ -dimensional examples.

Proof.

□

Problem 4

State the definition of a real Hilbert space and give both finite- and ∞ -dimensional examples.

Proof.

□

Problem 5

Determine whether $\{\cos kx : k = 0, 1, 2, \dots\}$ is an orthogonal set in $\mathcal{L}^2[0, \pi]$.

Proof.

□

Problem 6 (32.13, p.98)

State and prove Cauchy's inequality.

Proof.

□

Problem 7 (33.9, p.104)

State and prove Minkowski's inequality in $\mathcal{L}^2$.

Proof.

□

Problem 8 (p.113)

Define a trigonometric polynomial and a trigonometric series.

Proof.

□

Problem 9

State and prove Bessel's inequality for an orthonormal set in $\mathcal{L}^2[-\pi, \pi]$.

Proof.

□

Problem 10

State and prove the Riemann-Lebesgue lemma for the Fourier series of $f \in \mathcal{L}^2[-\pi, \pi]$.

Proof.

□

Problem 11 (37.4, p.121)

Give two formulas for the Dirichlet kernel $D_n(t)$.

Proof.

□

Problem 12 (37.5, p.123)

Express the partial sum $s_n(x)$ of the Fourier series for a function $f \in \mathcal{L}^2[-\pi, \pi]$ using the Lebesgue integral and the Dirichlet kernel.

Proof.

□

Problem 13 (37.6, p.124)

Give a formula for the Fejér kernel $K_n(t)$ and relate it to the Dirichlet kernel.

Proof.

□

Problem 14

Define the Cesàro $(C, 1)$ -sum of a series $\sum_{k=1}^{\infty} a_k$. Give an example of a divergent series with a convergent $(C, 1)$ -sum.

Proof.

□

Problem 15 (37.6, p.124)

Express the $(C, 1)$ -sum $\sigma_n(x)$ of the Fourier series for a function $f \in \mathcal{L}[-\pi, \pi]$ using the Lebesgue integral and the Fejér kernel.

Proof.

□

Problem 16 (37.8, p.125)

Prove that if $f \in \mathcal{L}^2[-\pi, \pi]$ and f is continuous, then $\{\sigma_n\}$ converges uniformly to f .

Proof.

□

Problem 17 (40.2, p.139)

Prove that if $f \in \mathcal{L}^2[-\pi, \pi]$ and f is differentiable at x , then $\{s_n(x)\}$ converges to $f(x)$.

Proof.

□

Problem 18 (41.2, p.141)

State and prove the Riemann localization lemma.

Proof.

□

Problem 19

Compute the complex exponential Fourier series of $f(x) = \sin x + 2 \cos 3x$.

Proof.

□